Tutorial 2

Problem 1

Let f denote an arbitrary function $f : \mathbb{R}^2 \to \mathbb{R}$. For each of the following statements, determine whether the statement is true or false. If false, provide an explanation or a counterexample.

(a) If $f(x_0, y_0) = f(x_1, y_1)$, then $x_0 = x_1$ and $y_0 = y_1$.

Solution: False. Counterexample: If we let $(x_0, y_0) = (1, 2)$, $(x_1, y_1) = (-1, -2)$ and $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto xy$, then $f(1, 2) = 2 = f(-1, -2)$, but $x_0 = 1 \neq -1 = x_1$ and $y_0 = 2 \neq -2 = y_1$.

(b) If $a \in \mathbb{R}$ then $f(ax, ay) = a^2 f(x, y)$.

Solution: False. Counterexample: For $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x + y$ and any $a \in \mathbb{R}$, we have $f(ax, ay) = ax + ay = a(x + y) = af(x, y)$. And if $a \notin \{0, 1\}$ and $f(x, y) \neq 0$ (which is e.g. the case if $x = y \neq 0$), then $af(x, y) \neq a^2 f(x, y)$.

(c) A vertical line (that is, a line parallel to the "z-axis") can cross the graph of f at most once.

Solution: True. By definition of a function, f maps each (x, y) to a unique real number z.

(d) Two level curves of f can intersect.

Solution: False. If this statement was true, then there would exist a point (x, y) that would be on two distinct level curves, say on the level curves of level C_1 and of level $C_2 (\neq C_1)$. That is, we would have $f(x, y) = C_1$ and $f(x, y) = C_2$. Contradiction, since $C_1 \neq C_2$.

Problem 2

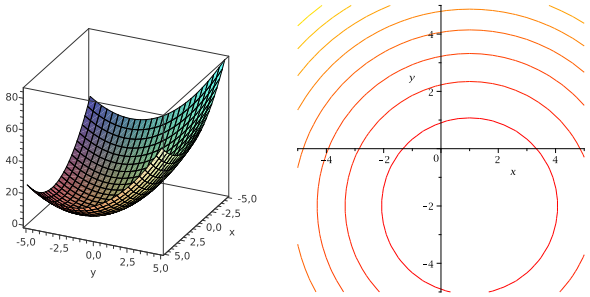
Recall that a *level curve* of a function (of two variables) f is a subset of f's domain whose elements are the points (x, y) for which for some $k \in \mathbb{R}$, $f(x, y) = k$. Determine the domain on which the following functions can be defined and draw a few level curves for each of them.

(a) $f(x, y) = (x - 1)^2 + (y + 2)^2$

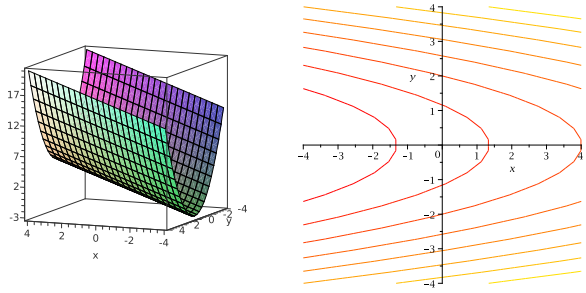
(b) $f(x, y) = x + y^2 + 1$

Solution: These three functions can be defined for all $(x, y) \in \mathbb{R}^2$.

(a) $f(x, y) = (x - 1)^2 + (y + 2)^2$



(b) $f(x, y) = x + y^2 + 1$
One level curve we determine here is for $k = 1$. Notice that $f(x, y) = 1$ is equivalent to $x + y^2 + 1 = 1$, which yields $x = -y^2$. This level curve has the same shape as those in the figure below and contains $(x, y) = (0, 0)$ so it goes through the origin.



Problem 3

Recall that $\ln(\cdot)$, $\sin(\cdot)$, and $\cos(\cdot)$ are continuous. Prove that the following functions are continuous.

(a) $f : \mathbb{R} \to \mathbb{R}, x \mapsto (\cos x)^2$

(b) $f : \mathbb{R}^2 \to \mathbb{R}^2, (x, y) \mapsto (\ln(x^4 + 1), \ln(y^4 + 1))$

(c) $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto \sin(x^2) + \sin(y^2)$

Solution: (Sketch only)

(a) $f = g \circ \cos$ for $g : \mathbb{R} \to \mathbb{R}, x \mapsto x^2$

(b) Each component function $f_i : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto \ln(x_i^4 + 1)$ of $f = (f_1, f_2)$ is continuous, as $f_i = \ln \circ g \circ pr_i$ for $g : \mathbb{R} \to \mathbb{R}, x \mapsto x^4 + 1$

(c) $f = \sin \circ g \circ pr_1 + \sin \circ g \circ pr_2$ for $g : \mathbb{R} \to \mathbb{R}, x \mapsto x^2$

Problem 4

The level curves of a production function $Q(K, L)$ are called *isoquants*. Obtain and plot a few isoquants for the following production functions.

(a) $Q(K, L) = K^{\frac{1}{2}} L^{\frac{1}{2}}$

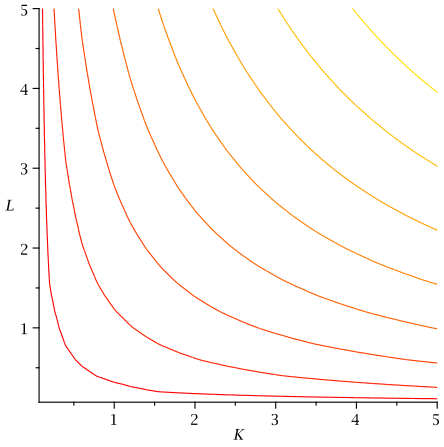
(b) $Q(K, L) = 0.5 \ln(K) + 0.5 \ln(L)$

Solution:

(a) $Q(K, L) = K^{\frac{1}{2}} L^{\frac{1}{2}}$

The level curve of level C is characterized by the equation $K^{\frac{1}{2}} L^{\frac{1}{2}} = C$, or, equivalently,

$$\begin{aligned} (K^{\frac{1}{2}} L^{\frac{1}{2}})^2 &= C^2, \\ KL &= C^2, \\ L &= \frac{C^2}{K} \end{aligned}$$



(b) $Q(K, L) = 0.5 \ln(K) + 0.5 \ln(L)$

The level curve of level C is characterized by the equation $0.5 \ln(K) + 0.5 \ln(L) = C$, or, equivalently,

$$\begin{aligned} e^{\ln(K^{\frac{1}{2}}) + \ln(L^{\frac{1}{2}})} &= e^C, \\ e^{\ln(K^{\frac{1}{2}})} e^{\ln(L^{\frac{1}{2}})} &= e^C, \\ K^{\frac{1}{2}} L^{\frac{1}{2}} &= e^C, \end{aligned}$$

from which it follows that the contour plot of this production function is identical to that in the figure for part (a).

Problem 5

(a) Write down the formula for a linear function g with domain $\mathbb{R}^2$ and codomain $\mathbb{R}$ that has the property that $g(7,2) - g(3,2) = 8$ and $g(7,5) - g(3,2) = 5$.

Solution: Since g is linear, we know there exists $\boldsymbol{a} = (a_1, a_2)$ such that $g(\boldsymbol{x}) = \boldsymbol{a} \cdot \boldsymbol{x} = a_1 x_1 + a_2 x_2$ for all $\boldsymbol{x} \in \mathbb{R}^2$. From

$$g(7,2) - g(3,2) = a_1 \cdot 7 + a_2 \cdot 2 - (a_1 \cdot 3 + a_2 \cdot 2) = 4a_1 = 8$$

we conclude that $a_1 = 2$. From

$$g(7,5) - g(3,2) = a_1 \cdot 7 + a_2 \cdot 5 - (a_1 \cdot 3 + a_2 \cdot 2) = 4a_1 + 3a_2 = 8 + 3a_2 = 5$$

we conclude that $a_2 = -1$. Hence $g : \mathbb{R}^2 \to \mathbb{R}, \boldsymbol{x} \mapsto (2,-1) \cdot \boldsymbol{x}$.

(b) Consider $\hat{g} : \mathbb{R}^2 \to \mathbb{R}, \boldsymbol{x} \mapsto g(\boldsymbol{x}) + 2$, where g is the linear function from (a). How does the value of $\hat{g}$ change if $\boldsymbol{x}$ changes

(i) from $(-3,0)$ to $(2,1)$?

(ii) from $(2,5)$ to $(1,3)$?

Solution: The change dy in the value of $\hat{g}$ if its variables change by $d\boldsymbol{x}$ is given by $dy = (2,-1) \cdot d\boldsymbol{x}$.

(i) Here, $d\boldsymbol{x} = (2,1) - (-3,0) = (5,1)$ and thus $dy = (2,-1) \cdot (5,1) = 10 - 1 = 9$.

(ii) Here, $d\boldsymbol{x} = (1,3) - (2,5) = (-1,-2)$ and thus $dy = (2,-1) \cdot (-1,-2) = -2 + 2 = 0$.

Problem 6 (Extra Practice Problem)

Let $A \subsetneq \mathbb{R}^k$. We say that a function $f : \mathbb{R}^k \backslash A \to \mathbb{R}$ can be **continuously extended** (to $\mathbb{R}^k$) if there is a continuous function $g : \mathbb{R}^k \to \mathbb{R}$ such that $f(\boldsymbol{x}) = g(\boldsymbol{x})$ for all $\boldsymbol{x} \in \mathbb{R}^k \backslash A$. Prove that the following functions cannot be continuously extended.[1]

(In the lectures, even though we did not use this language, we showed that $f : \mathbb{R}^2 \backslash \{(0,0)\} \to \mathbb{R}, (x,y) \mapsto \frac{xy}{x^2+y^2}$ cannot be continuously extended.)

(a) $f : \mathbb{R}^2 \backslash \{(0,0)\} \to \mathbb{R}, (x,y) \mapsto \frac{xy^2}{x^4+y^4}$
(Hint: Consider a sequence in the x-y-plane that approaches $(0,0)$ along the diagonal.)

[1] Part (a) is similar to SB, Exercise 13.21. Note that in SB, there is a typo: SB erroneously write x^2 instead of x^4 in the function - this is Part (b), and requires a different hint than the one given in SB!

(b) $f : \mathbb{R}^2 \backslash \{(0,0)\} \to \mathbb{R}, (x,y) \mapsto \frac{xy^2}{x^2+y^4}$
(Hint: Consider a sequence in the x-y-plane that approaches $(0,0)$ along a line, and a sequence that approaches $(0,0)$ along a certain parabola.)

Solution:

(a) Consider the sequence $\left(\frac{1}{n}, \frac{1}{n}\right)_{n\in\mathbb{N}}$. Since $\frac{1}{n} \to 0$, this sequence converges to $(0,0)$, and if g was a continuous extension of f, then we would have $f\left(\frac{1}{n}, \frac{1}{n}\right) \to g(0,0)$. However,

$$f\left(\frac{1}{n}, \frac{1}{n}\right) = \frac{\frac{1}{n}\left(\frac{1}{n}\right)^2}{\left(\frac{1}{n}\right)^4 + \left(\frac{1}{n}\right)^4} = \frac{n}{2},$$

implying that $\left(f\left(\frac{1}{n}, \frac{1}{n}\right)\right)_{n\in\mathbb{N}}$ diverges. Therefore, it is impossible to extend f continuously.

(b) Consider the sequences $\left(\frac{1}{n}, 0\right)_{n\in\mathbb{N}}$ and $\left(\frac{1}{n^2}, \frac{1}{n}\right)_{n\in\mathbb{N}}$. Both of these sequences converge to $(0,0)$ (the second sequence along the parabola $x = y^2$). However,

$$f\left(\frac{1}{n}, 0\right) = 0 \to 0$$

and

$$f\left(\frac{1}{n^2}, \frac{1}{n}\right) = \frac{\frac{1}{n^2}\left(\frac{1}{n}\right)^2}{\left(\frac{1}{n^2}\right)^2 + \left(\frac{1}{n}\right)^4} = \frac{1}{2} \to \frac{1}{2},$$

implying that no extension of f can be continuous.

This is a particularly striking example of how difficult it can be to prove that a function is continuous: at first, it may seem as if letting $f(0,0) = 0$ continuously extends f. This is because for *all* sequences $(\boldsymbol{x}_n)_{n\in\mathbb{N}}$ in $\mathbb{R}^2$ that approach $(0,0)$ *on a line*, we get $f(\boldsymbol{x}_n) \to 0$. (To see this, note that for $y = mx$, $m \in \mathbb{R}$, we have $f(x,y) = \frac{x(mx)^2}{x^2+(mx)^4} = \frac{m^2x}{1+m^4x^2}$. If $x \to 0$, we get $f(x,mx) \to \frac{0}{1+0} = 0$. The only line that is not of the form $y = mx$ is $x = 0$. In this case, $f(0,y) = 0$ for $y \neq 0$ and so if y approaches 0, $f(0,y) \to 0$.) However, continuity at $(0,0)$ requires that $f(\boldsymbol{x}_n) \to 0$ for all $(\boldsymbol{x}_n)_{n\in\mathbb{N}}$ converging to $(0,0)$, *even those who do not approach* $(0,0)$ *on a line!*

Problem 7 (Extra Practice Problem)

Determine whether the function $f : \mathbb{R}^2 \backslash \{(x_1,x_2) \in \mathbb{R}^2 : x_1 = x_2\} \to \mathbb{R}, (x_1,x_2) \mapsto \frac{x_1^2-x_2^2}{x_1-x_2}$ can be continuously extended to $\mathbb{R}^2$. If yes, show how. If no, prove why.

Solution: Yes, f can be continuously extended. Clearly,

$$g : \mathbb{R}^2 \to \mathbb{R}, (x_1,x_2) \mapsto \begin{cases} \frac{x_1^2-x_2^2}{x_1-x_2} & \text{if } x_1 \neq x_2 \\ x_1 + x_2 & \text{if } x_1 = x_2 \end{cases}$$

extends f to $\mathbb{R}^2$. Note that

$$\frac{x_1^2 - x_2^2}{x_1 - x_2} = \frac{(x_1 + x_2)(x_1 - x_2)}{x_1 - x_2} = x_1 + x_2,$$

and that therefore, we can simplify and write g as $g : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto x_1 + x_2$. Since $g = pr_1 + pr_2$, g is continuous. To sum up, g continuously extends f.